# CS : COMPUTER SCIENCE & INFORMATION TECHNOLOGY

*Duration*: Three Hours *Maximum Marks*: 100

**Read the following instructions carefully.**

1. Do not open the seal of the Question Booklet until you are asked to do so by the invigilator.
2. Take out the **O**ptical **R**esponse **S**heet (**ORS**) from this Question Booklet **without breaking the seal**. If you find that the Question Booklet Code printed at the right hand top corner of this page does not match with the Booklet Code on the **ORS**, exchange the booklet immediately with a new sealed Question Booklet.
3. Write your registration number, your name and name of the examination centre at the specified locations on the right half of the **ORS**. Also, using HB pencil, darken the appropriate bubble under each digit of your registration number and the letters corresponding to your test paper code (CS).
4. Write your name and registration number in the space provided at the bottom of this page.
5. This Booklet contains **20** pages including blank pages for rough work. After opening the seal at the specified time, please check all pages and report discrepancy, if any.
6. There are a total of 65 questions carrying 100 marks. All these questions are of objective type. Questions must be answered on the left hand side of the **ORS** by darkening the appropriate bubble (marked A, B, C, D) using HB pencil against the question number. **For each question darken the bubble of the correct answer**. In case you wish to change an answer, erase the old answer completely. More than one answer bubbled against a question will be treated as an incorrect response.
7. Questions Q.1 – Q.25 carry 1-mark each, and questions Q.26 – Q.55 carry 2-marks each.
8. Questions Q.48 – Q.51 (2 pairs) are common data questions and question pairs (Q.52, Q.53) and (Q.54, Q.55) are linked answer questions. The answer to the second question of the linked answer questions depends on the answer to the first question of the pair. If the first question in the linked pair is wrongly answered or is unattempted, then the answer to the second question in the pair will not be evaluated.
9. Questions Q.56 – Q.65 belong to General Aptitude (GA). Questions Q.56 – Q.60 carry 1-mark each, and questions Q.61 – Q.65 carry 2-marks each. The GA questions begin on a fresh page starting from page **16**.
10. Unattempted questions will result in zero mark and wrong answers will result in **NEGATIVE** marks. For Q.1 – Q.25 and Q.56 – Q.60, 1/3 mark will be deducted for each wrong answer. For Q.26 – Q.51 and Q.61 – Q.65, 2/3 mark will be deducted for each wrong answer. The question pairs (Q.52, Q.53), and (Q.54, Q.55) are questions with linked answers. There will be negative marks only for wrong answer to the first question of the linked answer question pair, i.e. for Q.52 and Q.54, 2/3 mark will be deducted for each wrong answer. There is no negative marking for Q.53 and Q.55.
11. Calculator is allowed whereas charts, graph sheets or tables are **NOT** allowed in the examination hall.
12. Rough work can be done on the question paper itself. Additionally, blank pages are provided at the end of the question paper for rough work.

| **Name** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Registration Number** | **CS** | | | | | | | |

## Q. 1 – Q. 25 carry one mark each.

Q.1 In a compiler, keywords of a language are recognized during

(A) parsing of the program
(B) the code generation
(C) the lexical analysis of the program
(D) dataflow analysis

Q.2 A layer-4 firewall (a device that can look at all protocol headers up to the transport layer) **CANNOT**

(A) block entire HTTP traffic during 9:00PM and 5:00AM
(B) block all ICMP traffic
(C) stop incoming traffic from a specific IP address but allow outgoing traffic to the same IP address
(D) block TCP traffic from a specific user on a multi-user system during 9:00PM and 5:00AM

Q.3 If two fair coins are flipped and at least one of the outcomes is known to be a head, what is the probability that both outcomes are heads?

(A) 1/3 (B) 1/4 (C) 1/2 (D) 2/3

Q.4 Consider different activities related to email.

m1: Send an email from a mail client to a mail server
m2: Download an email from mailbox server to a mail client
m3: Checking email in a web browser

Which is the application level protocol used in each activity?

(A) m1: HTTP m2: SMTP m3: POP
(B) m1: SMTP m2: FTP m3: HTTP
(C) m1: SMTP m2: POP m3: HTTP
(D) m1: POP m2: SMTP m3: IMAP

Q.5 A company needs to develop a strategy for software product development for which it has a choice of two programming languages L1 and L2. The number of lines of code (LOC) developed using L2 is estimated to be twice the LOC developed with L1. The product will have to be maintained for five years. Various parameters for the company are given in the table below.

| Parameter | Language L1 | Language L2 |
|---|---|---|
| Man years needed for development | LOC/10000 | LOC/10000 |
| Development Cost per man year | ₹ 10,00,000 | ₹ 7,50,000 |
| Maintenance time | 5 years | 5 years |
| Cost of maintenance per year | ₹ 1,00,000 | ₹ 50,000 |

Total cost of the project includes cost of development and maintenance. What is the LOC for L1 for which the cost of the project using L1 is equal to the cost of the project using L2?

(A) 4000 (B) 5000 (C) 4333 (D) 4667

Q.6 Let the time taken to switch between user and kernel modes of execution be $t_1$ while the time taken to switch between two processes be $t_2$. Which of the following is **TRUE**?

(A) $t_1 > t_2$
(B) $t_1 = t_2$
(C) $t_1 < t_2$
(D) nothing can be said about the relation between $t_1$ and $t_2$

Q.7 A company needs to develop digital signal processing software for one of its newest inventions. The software is expected to have 40000 lines of code. The company needs to determine the effort in person-months needed to develop this software using the basic COCOMO model. The multiplicative factor for this model is given as 2.8 for the software development on embedded systems, while the exponentiation factor is given as 1.20. What is the estimated effort in person -months?

(A) 234.25 (B) 932.50 (C) 287.80 (D) 122.40

Q.8 Which of the following pairs have **DIFFERENT** expressive power?

(A) Deterministic finite automata (DFA) and Non-deterministic finite automata (NFA)
(B) Deterministic push down automata (DPDA) and Non-deterministic push down automata (NPDA)
(C) Deterministic single-tape Turing machine and Non-deterministic single-tape Turing machine
(D) Single-tape Turing machine and multi-tape Turing machine

Q.9 HTML (HyperText Markup Language) has language elements which permit certain actions other than describing the structure of the web document. Which one of the following actions is **NOT** supported by pure HTML (without any server or client side scripting) pages?

(A) Embed web objects from different sites into the same page
(B) Refresh the page automatically after a specified interval
(C) Automatically redirect to another page upon download
(D) Display the client time as part of the page

Q.10 Which one of the following is **NOT** desired in a good Software Requirement Specifications (SRS) document?

(A) Functional Requirements (B) Non-Functional Requirements
(C) Goals of Implementation (D) Algorithms for Software Implementation

Q.11 A computer handles several interrupt sources of which the following are relevant for this question.

- Interrupt from CPU temperature sensor (raises interrupt if CPU temperature is too high)
- Interrupt from Mouse (raises interrupt if the mouse is moved or a button is pressed)
- Interrupt from Keyboard (raises interrupt when a key is pressed or released)
- Interrupt from Hard Disk (raises interrupt when a disk read is completed)

Which one of these will be handled at the **HIGHEST** priority?

(A) Interrupt from Hard Disk (B) Interrupt from Mouse
(C) Interrupt from Keyboard (D) Interrupt from CPU temperature sensor

Q.12 Consider a relational table with a single record for each registered student with the following attributes.

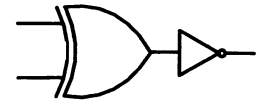
1. *Registration_Num*: Unique registration number of each registered student
2. *UID*: Unique identity number, unique at the national level for each citizen
3. *BankAccount_Num*: Unique account number at the bank. A student can have multiple accounts or joint accounts. This attribute stores the primary account number.
4. *Name*: Name of the student
5. *Hostel_Room*: Room number of the hostel

Which of the following options is **INCORRECT**?

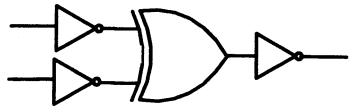
(A) *BankAccount_Num* is a candidate key
(B) *Registration_Num* can be a primary key
(C) *UID* is a candidate key if all students are from the same country
(D) If $S$ is a superkey such that $S \cap UID$ is NULL then $S \cup UID$ is also a superkey

Q.13 Which one of the following circuits is **NOT** equivalent to a 2-input XNOR (exclusive NOR) gate?

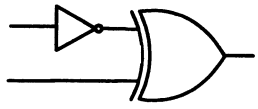
(A)



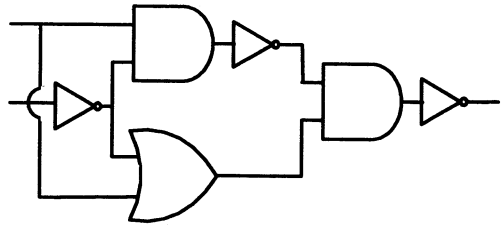
(B)



(C)



(D)



Q.14 The simplified SOP (Sum of Product) form of the Boolean expression $\left(P+\overline{Q}+\overline{R}\right).\left(P+\overline{Q}+R\right).\left(P+Q+\overline{R}\right)$ is

(A) $\left(\overline{P}.Q+\overline{R}\right)$ (B) $\left(P+\overline{Q}.\overline{R}\right)$
(C) $\left(\overline{P}.Q+R\right)$ (D) $\left(P.Q+R\right)$

Q.15 The minimum number of D flip-flops needed to design a mod-258 counter is

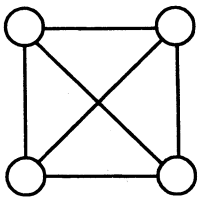
(A) 9 (B) 8 (C) 512 (D) 258

Q.16 A thread is usually defined as a "light weight process" because an operating system (OS) maintains smaller data structures for a thread than for a process. In relation to this, which of the following is **TRUE**?

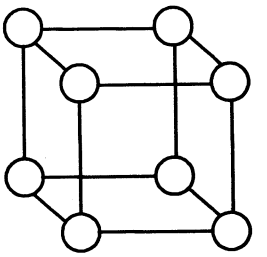
(A) On per-thread basis, the OS maintains *only* CPU register state
(B) The OS does not maintain a separate stack for each thread
(C) On per-thread basis, the OS does not maintain virtual memory state
(D) On per-thread basis, the OS maintains *only* scheduling and accounting information

Q.17 K4 and Q3 are graphs with the following structures.

K4



Q3



Which one of the following statements is **TRUE** in relation to these graphs?

(A) K4 is planar while Q3 is not
(B) Both K4 and Q3 are planar
(C) Q3 is planar while K3 is not
(D) Neither K4 nor Q3 is planar

Q.18 If the difference between the expectation of the square of a random variable $\left(E\left[X^2\right]\right)$ and the square of the expectation of the random variable $\left(E\left[X\right]\right)^2$ is denoted by $R$, then

(A) $R=0$ (B) $R<0$ (C) $R\geq 0$ (D) $R>0$

Q.19 The lexical analysis for a modern computer language such as Java needs the power of which one of the following machine models in a necessary and sufficient sense?

(A) Finite state automata
(B) Deterministic pushdown automata
(C) Non-deterministic pushdown automata
(D) Turing machine

Q.20 Let the page fault service time be 10 ms in a computer with average memory access time being 20 ns. If one page fault is generated for every $10^6$ memory accesses, what is the effective access time for the memory?

(A) 21 ns (B) 30 ns (C) 23 ns (D) 35 ns

Q.21 Consider a hypothetical processor with an instruction of type `LW R1, 20(R2)`, which during execution reads a 32-bit word from memory and stores it in a 32-bit register R1. The effective address of the memory location is obtained by the addition of a constant 20 and the contents of register R2. Which of the following best reflects the addressing mode implemented by this instruction for the operand in memory?

(A) Immediate Addressing
(B) Register Addressing
(C) Register Indirect Scaled Addressing
(D) Base Indexed Addressing

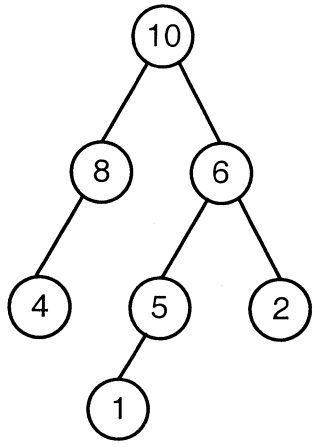
Q.22 What does the following fragment of C program print?

```
Char c[] = "GATE2011";
char *p  = c;
printf("%s", p + p[3] - p[1]);
```

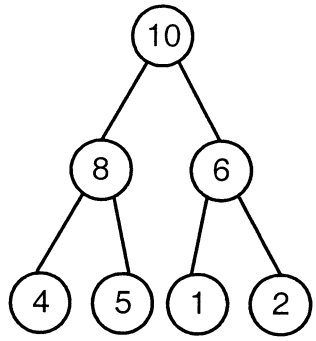
(A) `GATE2011` (B) `E2011` (C) `2011` (D) `011`

Q.23 A max-heap is a heap where the value of each parent is greater than or equal to the value of its children. Which of the following is a max-heap?

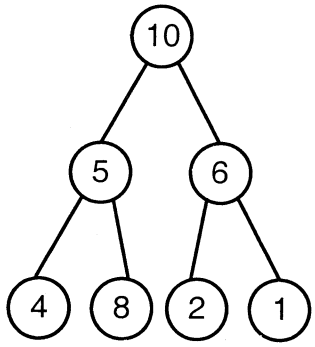
(A)



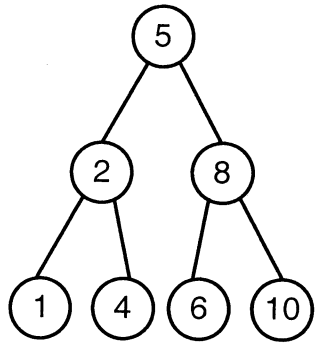

(B)




(C)




(D)




Q.24 Let $P$ be a regular language and $Q$ be a context-free language such that $Q \subseteq P$. (For example, let $P$ be the language represented by the regular expression $p^*q^*$ and $Q$ be $\{p^n q^n \mid n \in \mathbb{N}\}$). Then which of the following is **ALWAYS** regular?

(A) $P \cap Q$ (B) $P - Q$ (C) $\Sigma^* - P$ (D) $\Sigma^* - Q$

Q.25 An algorithm to find the length of the longest monotonically increasing sequence of numbers in an array $A[0:n-1]$ is given below.

Let $L_i$ denote the length of the longest monotonically increasing sequence starting at index $i$ in the array.

Initialize $L_{n-1} = 1$.

For all $i$ such that $0 \le i \le n-2$

$$L_i = \begin{cases} 1 + L_{i+1} & \text{if } A[i] < A[i+1] \\ 1 & \text{Otherwise} \end{cases}$$

Finally the length of the longest monotonically increasing sequence is $\text{Max}(L_0, L_1, \ldots, L_{n-1})$.

Which of the following statements is **TRUE**?

(A) The algorithm uses dynamic programming paradigm
(B) The algorithm has a linear complexity and uses branch and bound paradigm
(C) The algorithm has a non-linear polynomial complexity and uses branch and bound paradigm
(D) The algorithm uses divide and conquer paradigm.

## Q. 26 to Q. 55 carry two marks each.

Q.26 Consider the languages $L1$, $L2$ and $L3$ as given below.

$L1 = \{0^p 1^q \mid p, q \in \mathbb{N}\}$,

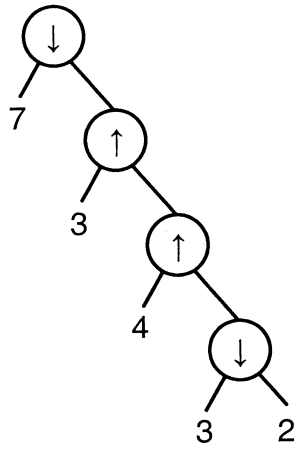
$L2 = \{0^p 1^q \mid p, q \in \mathbb{N} \text{ and } p = q\}$ and

$L3 = \{0^p 1^q 0^r \mid p, q, r \in \mathbb{N} \text{ and } p = q = r\}$. Which of the following statements is **NOT TRUE**?

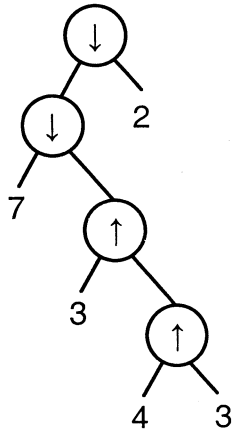
(A) Push Down Automata (PDA) can be used to recognize $L1$ and $L2$
(B) $L1$ is a regular language
(C) All the three languages are context free
(D) Turing machines can be used to recognize all the languages

Q.27 Consider two binary operators '↑' and '↓' with the precedence of operator ↓ being lower than that of the operator ↑. Operator ↑ is right associative while operator ↓ is left associative. Which one of the following represents the parse tree for expression (7↓3↑4↑3↓2)?

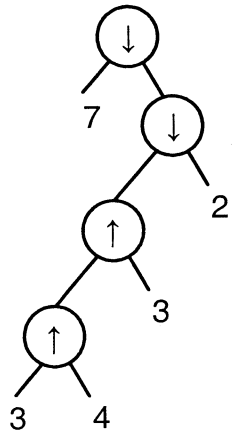
(A)



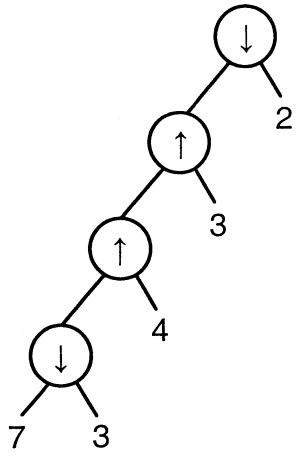

(B)




(C)




(D)



Q.28 On a non-pipelined sequential processor, a program segment, which is a part of the interrupt service routine, is given to transfer 500 bytes from an I/O device to memory.

```
        Initialize the address register
        Initialize the count to 500
LOOP:   Load a byte from device
        Store in memory at address given by address register
        Increment the address register
        Decrement the count
        If count != 0 go to LOOP
```

Assume that each statement in this program is equivalent to a machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load-store instructions take two clock cycles to execute.

The designer of the system also has an alternate approach of using the DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory.

What is the approximate speedup when the DMA controller based design is used in place of the interrupt driven program based input-output?

(A) 3.4 (B) 4.4 (C) 5.1 (D) 6.7

Q.29 We are given a set of $n$ distinct elements and an unlabeled binary tree with $n$ nodes. In how many ways can we populate the tree with the given set so that it becomes a binary search tree?

(A) 0 (B) 1 (C) $n!$ (D) $\frac{1}{n+1} \cdot {}^{2n}C_n$

Q.30 Which one of the following options is **CORRECT** given three positive integers $x$, $y$ and $z$, and a predicate

$$P(x) = \neg(x=1) \wedge \forall y\left(\exists z(x = y * z) \Rightarrow (y = x) \vee (y = 1)\right)$$

(A) $P(x)$ being true means that $x$ is a prime number

(B) $P(x)$ being true means that $x$ is a number other than 1

(C) $P(x)$ is always true irrespective of the value of $x$

(D) $P(x)$ being true means that $x$ has exactly two factors other than 1 and $x$

Q.31 Given $i = \sqrt{-1}$, what will be the evaluation of the definite integral $\int_0^{\pi/2} \frac{\cos x + i \sin x}{\cos x - i \sin x}\, dx$?

(A) 0 (B) 2 (C) $-i$ (D) $i$

Q.32 Consider a database table T containing two columns X and Y each of type `integer`. After the creation of the table, one record (X=1, Y=1) is inserted in the table.

Let MX and MY denote the respective maximum values of X and Y among all records in the table at any point in time. Using MX and MY, new records are inserted in the table 128 times with X and Y values being MX+1, 2*MY+1 respectively. It may be noted that each time after the insertion, values of MX and MY change.

What will be the output of the following SQL query after the steps mentioned above are carried out?

SELECT Y FROM T WHERE X=7;

(A) 127 (B) 255 (C) 129 (D) 257

Q.33 Consider a finite sequence of random values $X = [x_1, x_2, \ldots, x_n]$. Let $\mu_x$ be the mean and $\sigma_x$ be the standard deviation of $X$. Let another finite sequence $Y$ of equal length be derived from this as $y_i = a * x_i + b$, where $a$ and $b$ are positive constants. Let $\mu_y$ be the mean and $\sigma_y$ be the standard deviation of this sequence. Which one of the following statements is **INCORRECT**?

(A) Index position of mode of $X$ in $X$ is the same as the index position of mode of $Y$ in $Y$.

(B) Index position of median of $X$ in $X$ is the same as the index position of median of $Y$ in $Y$.

(C) $\mu_y = a\mu_x + b$

(D) $\sigma_y = a\sigma_x + b$

Q.34 A deck of 5 cards (each carrying a distinct number from 1 to 5) is shuffled thoroughly. Two cards are then removed one at a time from the deck. What is the probability that the two cards are selected with the number on the first card being one higher than the number on the second card?

(A) 1/5 (B) 4/25 (C) 1/4 (D) 2/5

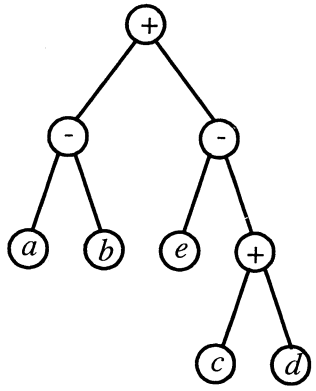
Q.35 Consider the following table of arrival time and burst time for three processes P0, P1 and P2.

| Process | Arrival time | Burst Time |
|---|---|---|
| P0 | 0 ms | 9 ms |
| P1 | 1 ms | 4 ms |
| P2 | 2 ms | 9 ms |

The pre-emptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of processes. What is the average waiting time for the three processes?

(A) 5.0 ms (B) 4.33 ms (C) 6.33 ms (D) 7.33 ms

Q.36 Consider evaluating the following expression tree on a machine with load-store architecture in which memory can be accessed only through load and store instructions. The variables $a$, $b$, $c$, $d$ and $e$ are initially stored in memory. The binary operators used in this expression tree can be evaluated by the machine only when the operands are in registers. The instructions produce result only in a register. If no intermediate results can be stored in memory, what is the minimum number of registers needed to evaluate this expression?




(A) 2 (B) 9 (C) 5 (D) 3

Q.37 Which of the given options provides the increasing order of asymptotic complexity of functions $f_1, f_2, f_3$ and $f_4$ ?

$f_1(n) = 2^n$ $f_2(n) = n^{3/2}$ $f_3(n) = n \log_2 n$ $f_4(n) = n^{\log_2 n}$

(A) $f_3, f_2, f_4, f_1$
(B) $f_3, f_2, f_1, f_4$
(C) $f_2, f_3, f_1, f_4$
(D) $f_2, f_3, f_4, f_1$

Q.38 Four matrices $M_1$, $M_2$, $M_3$ and $M_4$ of dimensions $p \times q$, $q \times r$, $r \times s$ and $s \times t$ respectively can be multiplied in several ways with different number of total scalar multiplications. For example when multiplied as $((M_1 \times M_2) \times (M_3 \times M_4))$, the total number of scalar multiplications is $pqr + rst + prt$. When multiplied as $(((M_1 \times M_2) \times M_3) \times M_4)$, the total number of scalar multiplications is $pqr + prs + pst$.

If $p = 10$, $q = 100$, $r = 20$, $s = 5$, and $t = 80$, then the minimum number of scalar multiplications needed is

(A) 248000 (B) 44000 (C) 19000 (D) 25000

Q.39 Consider a relational table $r$ with sufficient number of records, having attributes $A_1, A_2, \ldots, A_n$ and let $1 \le p \le n$. Two queries Q1 and Q2 are given below.

Q1: $\pi_{A_1,\ldots,A_p}\left(\sigma_{A_p=c}(r)\right)$ where $c$ is a constant

Q2: $\pi_{A_1,\ldots,A_p}\left(\sigma_{c_1 \le A_p \le c_2}(r)\right)$ where $c_1$ and $c_2$ are constants

The database can be configured to do ordered indexing on $A_p$ or hashing on $A_p$. Which of the following statements is **TRUE**?

(A) Ordered indexing will always outperform hashing for both queries
(B) Hashing will always outperform ordered indexing for both queries
(C) Hashing will outperform ordered indexing on Q1, but not on Q2
(D) Hashing will outperform ordered indexing on Q2, but not on Q1

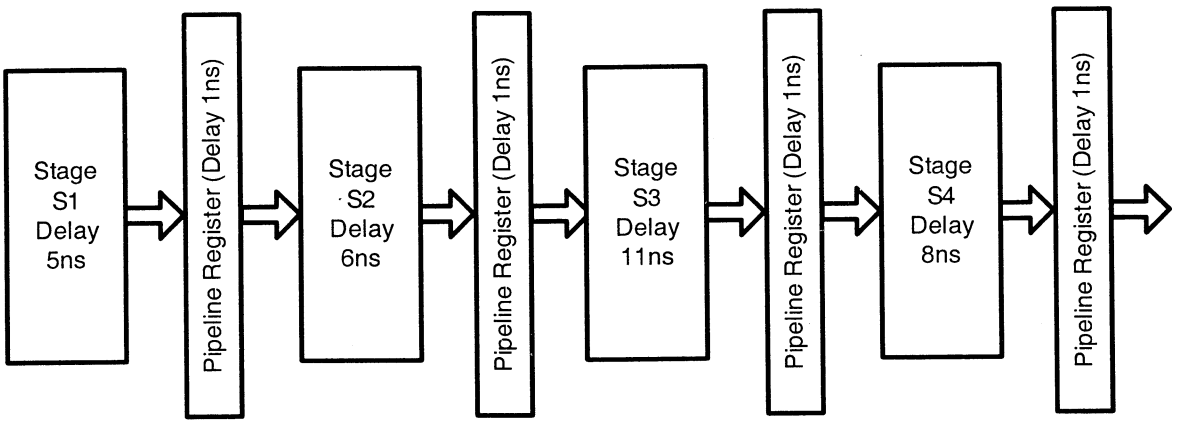
Q.40 Consider the matrix as given below.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 7 \\ 0 & 0 & 3 \end{bmatrix}$$

Which one of the following options provides the **CORRECT** values of the eigenvalues of the matrix?

(A) 1, 4, 3 (B) 3, 7, 3 (C) 7, 3, 2 (D) 1, 2, 3

Q.41 Consider an instruction pipeline with four stages (S1, S2, S3 and S4) each with combinational circuit only. The pipeline registers are required between each stage and at the end of the last stage. Delays for the stages and for the pipeline registers are as given in the figure.




What is the approximate speed up of the pipeline in steady state under ideal conditions when compared to the corresponding non-pipeline implementation?

(A) 4.0 (B) 2.5 (C) 1.1 (D) 3.0

Q.42 Definition of a language $L$ with alphabet $\{a\}$ is given as following.

$$L = \{a^{nk} \mid k > 0, \text{ and } n \text{ is a positive integer constant}\}$$

What is the minimum number of states needed in a DFA to recognize $L$?

(A) $k+1$ (B) $n+1$ (C) $2^{n+1}$ (D) $2^{k+1}$

Q.43 An 8KB direct-mapped write-back cache is organized as multiple blocks, each of size 32-bytes. The processor generates 32-bit addresses. The cache controller maintains the tag information for each cache block comprising of the following.

1 Valid bit
1 Modified bit
As many bits as the minimum needed to identify the memory block mapped in the cache.

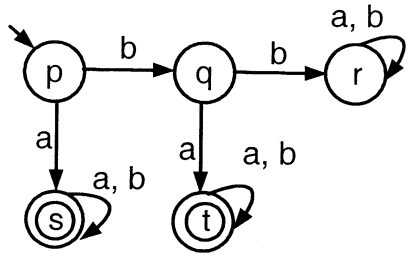
What is the total size of memory needed at the cache controller to store meta-data (tags) for the cache?

(A) 4864 bits (B) 6144 bits (C) 6656 bits (D) 5376 bits

Q.44 An application loads 100 libraries at startup. Loading each library requires exactly one disk access. The seek time of the disk to a random location is given as 10 ms. Rotational speed of disk is 6000 rpm. If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected.)

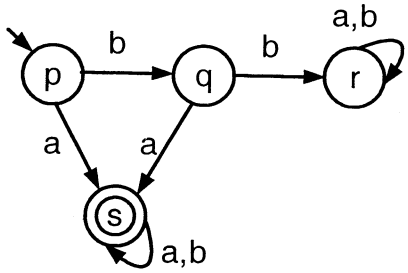
(A) 0.50 s (B) 1.50 s (C) 1.25 s (D) 1.00 s

Q.45 A deterministic finite automaton (DFA) $D$ with alphabet $\sum = \{a, b\}$ is given below.

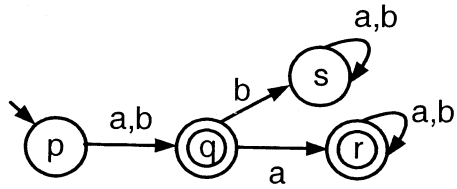



Which of the following finite state machines is a valid minimal DFA which accepts the same language as $D$?

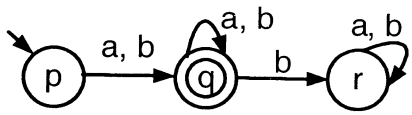
(A)



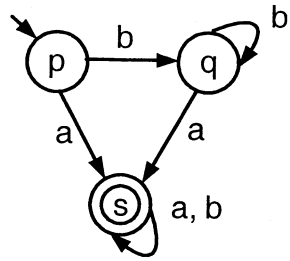

(B)




(C)




(D)




Q.46 Database table by name `Loan_Records` is given below.

| Borrower | Bank_Manager | Loan_Amount |
|---|---|---|
| Ramesh | Sunderajan | 10000.00 |
| Suresh | Ramgopal | 5000.00 |
| Mahesh | Sunderajan | 7000.00 |

What is the output of the following SQL query?

```
SELECT count(*)
FROM (
        (SELECT Borrower, Bank_Manager FROM Loan_Records) AS S
        NATURAL JOIN
        (SELECT Bank_Manager, Loan_Amount FROM Loan_Records) AS T
);
```

(A) 3 (B) 9 (C) 5 (D) 6

Q.47 The following is the comment written for a C function.

```
/* This function computes the roots of a quadratic equation
   a.x^2 + b.x + c = 0. The function stores two real roots
   in *root1 and *root2 and returns the status of validity
   of roots. It handles four different kinds of cases.
   (i) When coefficient a is zero irrespective of discriminant
   (ii) When discriminant is positive
   (iii) When discriminant is zero
   (iv) When discriminant is negative.
   Only in case (ii) and (iii), the stored roots are valid.
   Otherwise 0 is stored in the roots. The function returns
   0 when the roots are valid and -1 otherwise.
   The function also ensures root1 >= root2.
      int get_QuadRoots(float a, float b, float c,
         float *root1, float *root2);
*/
```

A software test engineer is assigned the job of doing black box testing. He comes up with the following test cases, many of which are redundant.

| Test Case | Input Set | | | Expected Output Set | | |
|---|---|---|---|---|---|---|
| | a | b | c | root1 | root2 | Return Value |
| T1 | 0.0 | 0.0 | 7.0 | 0.0 | 0.0 | −1 |
| T2 | 0.0 | 1.0 | 3.0 | 0.0 | 0.0 | −1 |
| T3 | 1.0 | 2.0 | 1.0 | −1.0 | −1.0 | 0 |
| T4 | 4.0 | −12.0 | 9.0 | 1.5 | 1.5 | 0 |
| T5 | 1.0 | −2.0 | −3.0 | 3.0 | −1.0 | 0 |
| T6 | 1.0 | 1.0 | 4.0 | 0.0 | 0.0 | −1 |

Which one of the following options provide the set of non-redundant tests using equivalence class partitioning approach from input perspective for black box testing?

(A) T1, T2, T3, T6
(B) T1, T3, T4, T5
(C) T2, T4, T5, T6
(D) T2, T3, T4, T5

## Common Data Questions

**Common Data for Questions 48 and 49:**

Consider the following recursive C function that takes two arguments.

```
unsigned int foo(unsigned int n, unsigned int r) {
    if (n>0) return ((n%r) + foo(n/r, r));
    else return 0;
}
```

Q.48 What is the return value of the function `foo` when it is called as `foo(345, 10)`?

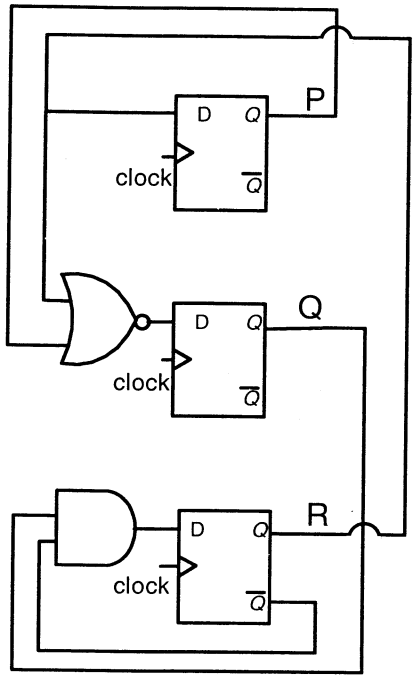
(A) 345 (B) 12 (C) 5 (D) 3

Q.49 What is the return value of the function `foo` when it is called as `foo(513, 2)`?

(A) 9 (B) 8 (C) 5 (D) 2

**Common Data for Questions 50 and 51:**

Consider the following circuit involving three D-type flip-flops used in a certain type of counter configuration.




Q.50 If at some instance prior to the occurrence of the clock edge, $P$, $Q$ and $R$ have a value 0, 1 and 0 respectively, what shall be the value of $PQR$ after the clock edge?

(A) 000 (B) 001 (C) 010 (D) 011

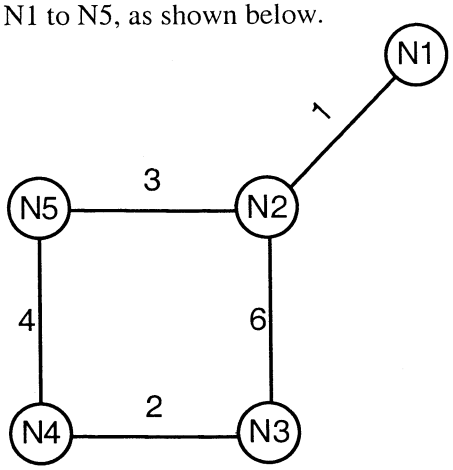
Q.51 If all the flip-flops were reset to 0 at power on, what is the total number of distinct outputs (states) represented by $PQR$ generated by the counter?

(A) 3 (B) 4 (C) 5 (D) 6

## Linked Answer Questions

**Statement for Linked Answer Questions 52 and 53:**

Consider a network with five nodes, N1 to N5, as shown below.




The network uses a Distance Vector Routing protocol. Once the routes have stabilized, the distance vectors at different nodes are as following.

N1: (0, 1, 7, 8, 4)
N2: (1, 0, 6, 7, 3)
N3: (7, 6, 0, 2, 6)
N4: (8, 7, 2, 0, 4)
N5: (4, 3, 6, 4, 0)

Each distance vector is the distance of the best known path at that instance to nodes, N1 to N5, where the distance to itself is 0. Also, all links are symmetric and the cost is identical in both directions. In each round, all nodes exchange their distance vectors with their respective neighbors. Then all nodes update their distance vectors. In between two rounds, any change in cost of a link will cause the two incident nodes to change only that entry in their distance vectors.

Q.52 The cost of link N2-N3 reduces to 2 (in both directions). After the next round of updates, what will be the new distance vector at node, N3?

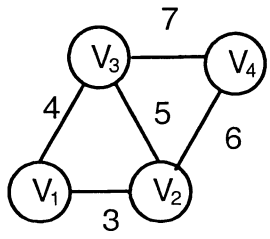
(A) (3, 2, 0, 2, 5) (B) (3, 2, 0, 2, 6) (C) (7, 2, 0, 2, 5) (D) (7, 2, 0, 2, 6)

Q.53 After the update in the previous question, the link N1-N2 goes down. N2 will reflect this change immediately in its distance vector as cost, $\infty$. After the **NEXT ROUND** of update, what will be the cost to N1 in the distance vector of N3?

(A) 3 (B) 9 (C) 10 (D) $\infty$

**Statement for Linked Answer Questions 54 and 55:**

An undirected graph $G(V,E)$ contains $n$ $(n>2)$ nodes named $v_1, v_2, \ldots, v_n$. Two nodes $v_i, v_j$ are connected if and only if $0 < |i-j| \leq 2$. Each edge $(v_i, v_j)$ is assigned a weight $i+j$. A sample graph with $n = 4$ is shown below.




Q.54 What will be the cost of the minimum spanning tree (MST) of such a graph with $n$ nodes?

(A) $\frac{1}{12}(11n^2 - 5n)$ (B) $n^2 - n + 1$ (C) $6n - 11$ (D) $2n + 1$

Q.55 The length of the path from $v_5$ to $v_6$ in the MST of previous question with $n = 10$ is

(A) 11 (B) 25 (C) 31 (D) 41

## General Aptitude (GA) Questions

## Q. 56 – Q. 60 carry one mark each.

Q.56 Which of the following options is the closest in the meaning to the word below:
**Inexplicable**

(A) Incomprehensible
(B) Indelible
(C) Inextricable
(D) Infallible

Q.57 If Log (P) = (1/2)Log (Q) = (1/3) Log (R), then which of the following options is **TRUE**?

(A) $P^2 = Q^3R^2$ (B) $Q^2 = PR$ (C) $Q^2 = R^3P$ (D) $R = P^2Q^2$

Q.58 Choose the most appropriate word(s) from the options given below to complete the following sentence.
**I contemplated _____ Singapore for my vacation but decided against it.**

(A) to visit
(B) having to visit
(C) visiting
(D) for a visit

Q.59 Choose the most appropriate word from the options given below to complete the following sentence.
**If you are trying to make a strong impression on your audience, you cannot do so by being understated, tentative or _____.**

(A) hyperbolic
(B) restrained
(C) argumentative
(D) indifferent

Q.60 Choose the word from the options given below that is most nearly opposite in meaning to the given word:
**Amalgamate**

(A) merge
(B) split
(C) collect
(D) separate

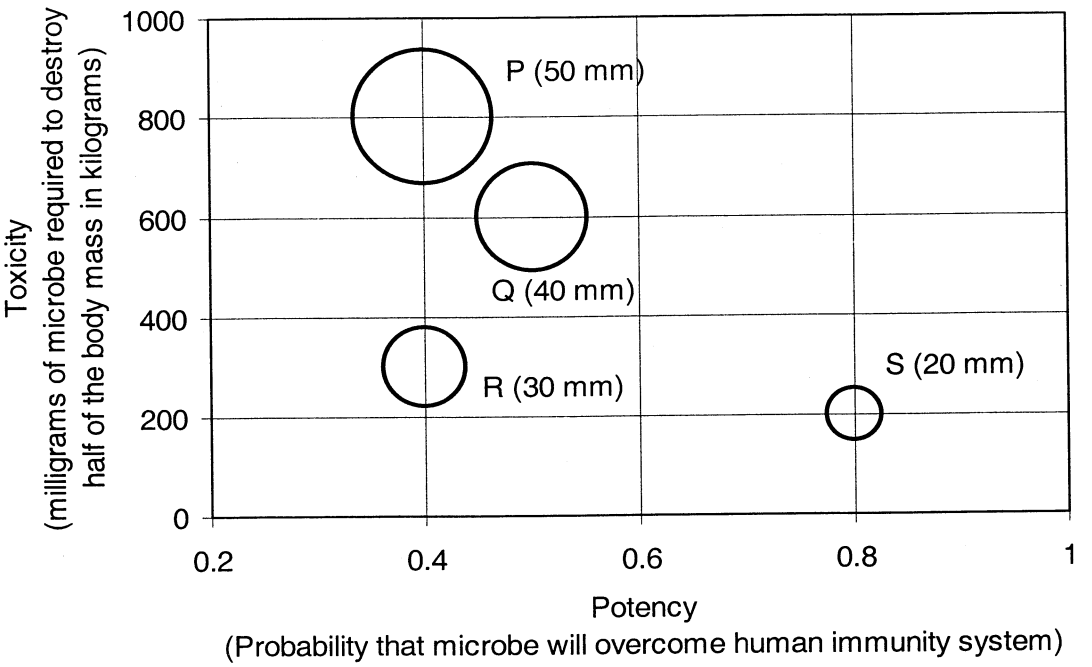
## Q. 61 to Q. 65 carry two marks each.

Q.61 **Few school curricula include a unit on how to deal with bereavement and grief, and yet all students at some point in their lives suffer from losses through death and parting.**

Based on the above passage which topic would not be included in a unit on bereavement?

(A) how to write a letter of condolence
(B) what emotional stages are passed through in the healing process
(C) what the leading causes of death are
(D) how to give support to a grieving friend

Q.62 P, Q, R and S are four types of dangerous microbes recently found in a human habitat. The area of each circle with its diameter printed in brackets represents the growth of a single microbe surviving human immunity system within 24 hours of entering the body. The danger to human beings varies proportionately with the toxicity, potency and growth attributed to a microbe shown in the figure below:




A pharmaceutical company is contemplating the development of a vaccine against the most dangerous microbe. Which microbe should the company target in its first attempt?

(A) P (B) Q (C) R (D) S

Q.63 The variable cost (V) of manufacturing a product varies according to the equation V= 4q, where q is the quantity produced. The fixed cost (F) of production of same product reduces with q according to the equation F = 100/q. How many units should be produced to minimize the total cost (V+F)?

(A) 5 (B) 4 (C) 7 (D) 6

Q.64 A transporter receives the same number of orders each day. Currently, he has some pending orders (backlog) to be shipped. If he uses 7 trucks, then at the end of the 4th day he can clear all the orders. Alternatively, if he uses only 3 trucks, then all the orders are cleared at the end of the 10th day. What is the minimum number of trucks required so that there will be no pending order at the end of the 5th day?

(A) 4 (B) 5 (C) 6 (D) 7

Q.65 A container originally contains 10 litres of pure spirit. From this container 1 litre of spirit is replaced with 1 litre of water. Subsequently, 1 litre of the mixture is again replaced with 1 litre of water and this process is repeated one more time. How much spirit is now left in the container?

(A) 7.58 litres (B) 7.84 litres (C) 7 litres (D) 7.29 litres

## END OF THE QUESTION PAPER